

Abstract Integration

1.1 The Concept of Measurability

Definition 1.1.1

1. A collection $\mathfrak{M}$ of subsets of a set X is said to be a σ -**algebra** in X if $\mathfrak{M}$ has the following properties:
 - (a) $X \in \mathfrak{M}$.
 - (b) If $A \in \mathfrak{M}$, then $A^c \in \mathfrak{M}$.
 - (c) If $A = \bigcup_{n=1}^{\infty} A_n$ and $A_n \in \mathfrak{M}$ for $n = 1, 2, 3, \dots$, then $A \in \mathfrak{M}$.
2. If $\mathfrak{M}$ is a σ -algebra in X , then X is called a **measurable** provided that $f^{-1}(V)$ is measurable set in X for every open set V in Y .
3. If X is a measurable space, Y is a topological space, and f is a mapping of X into Y , then f is said to be **measurable** provided that $f^{-1}(V)$ is a measurable set in X for every open set V in Y .

Note.

对比 topological space 的定义, σ -algebra 的定义是对称的, 即可测集的补集仍是可测的, 而对于拓扑空间, 开集的补集不是开集, 而是被定义为了闭集这样的对象.

Lemma 1.1.2

1. Let X be a measurable space, Y, Z be topological spaces. If $f : X \rightarrow Y$ is measurable, and if $g : Y \rightarrow Z$ is continuous, then $g \circ f : X \rightarrow Z$ is measurable.
2. Let u, v be real measurable functions on a measurable space X , let Φ be a continuous mapping of the plane into a topological space Y , and define

$$h(x) = \Phi(u(x), v(x))$$

for $x \in X$. Then $h : X \rightarrow Y$ is measurable.

Corollary 1.1.3

Let X be a measurable space. The following propositions hold

1. If $f = u + iv$, where u and v are real measurable functions on X , then f is a complex measurable function on X .
2. If $f = u + iv$ is a complex measurable function on X , then u , v , and $|f|$ are real measurable functions on X .
3. If f and g are complex measurable functions on X , then so are $f + g$ and fg .
4. If E is a measurable set in X and if

$$\chi_E(x) = \begin{cases} 1 & x \in E \\ 0 & x \notin E \end{cases}$$

then χ_E is a measurable function.

5. If f is a complex measurable function on X , there is a complex measurable function α on X such that $|\alpha| = 1$ and $f = \alpha |f|$.

Theorem 1.1.4

If $\mathcal{F}$ is any collection of subsets of X , there exists a smallest σ -algebra $\mathfrak{M}^*$ such that $\mathcal{F} \subseteq \mathfrak{M}^*$.

Definition 1.1.5 ▶ Borel

Let X be a topological space.

1. By **Borel σ -algebra**, we mean the smallest σ -algebra $\mathcal{B}$ in X such that every open set in X belongs to $\mathcal{B}$. The members of $\mathcal{B}$ are called the **Borel sets** of X .
2. All countable unions of closed sets and all countable intersections of open sets are Borel sets, which we called F_σ 's **and** G_δ 's, respectively.
3. By a **Borel function**, we mean a measurable function on the measurable space $(X, \mathcal{B})$.

Remark.

1. The letters F and G were used for closed and open sets, respectively, and σ refers to union, δ to intersection.
2. A continuous function is Borel-measurable, since the preimage of any open set is open and therefore a Borel set.

Theorem 1.1.6

Suppose $\mathfrak{M}$ is a σ -algebra in X , and Y is a topological space. Let f map X into Y .

1. If Ω is the collection of all sets $E \subseteq Y$ such that $f^{-1}(E) \in \mathfrak{M}$, then Ω is a σ -algebra in Y .
2. If f is measurable and E is a Borel set in Y , then $f^{-1}(E) \in \mathfrak{M}$.
3. If $Y = [-\infty, \infty]$ and $f^{-1}((\alpha, \infty]) \in \mathfrak{M}$ for every real α , then f is measurable.
4. If f is measurable, if Z is topological space, if $g : Y \rightarrow Z$ is a Borel mapping, and if $h = g \circ f$, then $h : X \rightarrow Z$ is measurable.

Theorem 1.1.7

If $f_n : X \rightarrow [-\infty, \infty]$ is measurable, for $n = 1, 2, 3, \dots$, and

$$g = \sup_{n \geq 1} f_n, \quad h = \lim_{n \rightarrow \infty} \sup f_n,$$

then g and h are measurable.

Corollary 1.1.8

1. The limit of every pointwise convergent sequence of complex measurable functions is measurable.
2. If f and g are measurable (with range in $[-\infty, \infty]$), then so are $\max\{f, g\}$ and $\min\{f, g\}$. In particular, this is true of the functions

$$f^+ = \max\{f, 0\}, \quad f^- = -\min\{f, 0\}.$$

which are called the **positive part** and **negative part** of f , respectively.

Remark.

1. There are the standard representation

$$|f| = f^+ + f^-, \quad f = f^+ - f^-$$

2. And easy but (may) useful observation is: If $f = g - h$, $g \geq 0$, $h \geq 0$, then $f^+ \leq g$ and $f^- \leq h$.

1.2 Simple Functions

Definition 1.2.1

A complex function s on a measurable space X whose range consists of only finitely many points will be called a **simple function**. Among these are the nonnegative simple functions, whose range is a finite subset of $[0, \infty)$.

Specifically, if $\alpha_1, \dots, \alpha_n$ are distinct values of a simple function s , and if we set $A_i = \{x : s(x) = \alpha_i\}$, then clearly

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i}.$$

Where χ_{A_i} is the characteristic function of A_i .

Remark.

- Here, we explicitly exclude ∞ from the values of a simple function.
- It is clear that s is measurable if and only if each of the sets A_i is measurable.

Theorem 1.2.2

Let $f : X \rightarrow [0, \infty]$ be measurable space. There exists simple measurable functions s_n on X such that

1. $0 \leq s_1 \leq s_2 \leq \dots \leq f$.
2. $s_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, for every $x \in X$.

Proof sketch.

We construct a sequence of Borel simple functions $\{\varphi_n(x)\}$ to act as an **identity**. 当 n 增大的同时, 我们同时让 $\varphi_n(x)$ 的单位逼近精度和单位逼近范围随着 n 提升. 并且在舍弃误差时, 总是向下取整, 使得该单位逼近是自下而上的.

Proof. For every $x \in [0, \infty]$, and for every $n \in \mathbb{N}$, there exists a unique integer $k_n(x)$, such that

$$k_n(x) 2^{-n} \leq x < (k_n(x) + 1) 2^{-n}$$

For every $n \in \mathbb{N}$, we define

$$\varphi_n(x) = \begin{cases} k_n(x) 2^{-n}, & 0 \leq x \leq n \\ n, & x \geq n \end{cases}$$

Each $\varphi_n(x)$ is then a Borel simple function. It is not hard to show that $0 \leq \varphi_1 \leq \varphi_2 \leq \dots \leq \text{Id}$. For each n , we define

$$s_n(x) := (\varphi_n \circ f)(x)$$

Then $\{s_n\}$ is a suquence of simple measurable functions such that $0 \leq s_1 \leq s_2 \leq \dots \leq f$. Since $\lim_{n \rightarrow \infty} \varphi_n(x) = x$, then $\lim_{n \rightarrow \infty} s_n(x) = f(x)$. $\square$

1.3 Measure

Definition 1.3.1 ► Measure and Measure Space

- (a) A **positive measure** is a function μ , defined on a σ -algebra $\mathfrak{M}$, whose range is in $[0, \infty]$ and which is **countably additive**. This means that if $\{A_i\}$ is a disjoint countable collection of members of $\mathfrak{M}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mu(A_i).$$

To avoid trivialities, we shall also assume that $\mu(A) < \infty$ for at least one $A \in \mathfrak{M}$.

- (b) A **measure space** is a measurable space which has a positive measure defined on the σ -algebra of its measurable sets.
- (c) A **complex measure** is a complex-valued countably additive function defined on a σ -algebra.

Theorem 1.3.2

Let μ be a positive measure on a σ -algebra $\mathfrak{M}$. Then

1. $\mu(\emptyset) = 0$.
2. $\mu(A_1 \cup \dots \cup A_n) = \mu(A_1) + \dots + \mu(A_n)$ if $A_1, \dots, A_n$ are pairwise disjoint members of $\mathfrak{M}$.
3. $A \subset B$ implies $\mu(A) \leq \mu(B)$ if $A \in \mathfrak{M}, B \in \mathfrak{M}$.

4. $\mu(A_n) \rightarrow \mu(A)$ as $n \rightarrow \infty$ if $A = \bigcup_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$, and

$$A_1 \subset A_2 \subset A_3 \subset \dots$$

5. $\mu(A_n) \rightarrow \mu(A)$ as $n \rightarrow \infty$ if $A = \bigcap_{n=1}^{\infty} A_n, A_n \in \mathfrak{M}$,

$$A_1 \supset A_2 \supset A_3 \supset \dots,$$

and $\mu(A_1)$ is finite.

Proof. 1. Take $A \in \mathfrak{M}$ such that $\mu(A) < \infty$.¹ And let $A_2 = A_3 = \dots = \emptyset$, then $\mu(\emptyset) > 0$ leads to a contradiction to the countably additive.

2. Take $A_{n+1} = A_{n+2} = \dots = \emptyset$.

3. Note that $B = (B \setminus A) \cup A$, then by additivity

$$\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$$

4. Let $A_0 = \emptyset$, and let $B_n = A_n \setminus A_{n-1}$ for all $n \in \mathbb{N}$. Then $B_1, \dots, B_n$ are pairwise disjoint members of $\mathfrak{M}$ such that $A = \bigcup_{n=1}^{\infty} B_n$. We have

$$\mu(A) = \sum_{n=1}^{\infty} \mu(B_n) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1})$$

If one of the $\mu(A_n)$ is ∞ , then $\lim_{n \rightarrow \infty} \mu(A_n)$ and $\mu(A)$ both are ∞ . Otherwise, we have

$$\mu(A_n \setminus A_{n-1}) = \mu(A_n) - \mu(A_{n-1}), \quad \forall n \in \mathbb{N}$$

Thus

$$\mu(A) = \sum_{n=1}^{\infty} \mu(A_n \setminus A_{n-1}) = \lim_{n \rightarrow \infty} \mu(A_n)$$

5. Let $B_n = A_n \setminus A_{n+1}$ for all $n \in \mathbb{N}$. $B_1, \dots, B_n$ are pairwise disjoint members of $\mathfrak{M}$ with finite measure, such that

$$A_1 \setminus A = A_1 \setminus \left(\bigcap_{n=1}^{\infty} A_n \right) = \bigcup_{n=1}^{\infty} (A_1 \setminus A_n) = \bigcup_{n=1}^{\infty} \left(\bigcup_{k=1}^n B_k \right) = \bigcup_{n=1}^{\infty} B_n$$

¹That is what we supposed at the definition of measure.

. We have

$$\mu(A_1 \setminus A) = \mu\left(\bigcup_{n=1}^{\infty} B_n\right),$$

where the RHS is $\mu(A_1) - \mu(A)$, and the LHS is

$$\sum_{n=1}^{\infty} (\mu(A_n) - \mu(A_{n+1})) = \mu(A_1) - \lim_{n \rightarrow \infty} \mu(A_{n+1})$$

Since $\mu(A_1) < \infty$, we have

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$$

□

Example 1.3.3

1. For any $E \subset X$, where X is any set, define $\mu(E) = \infty$ if E is an infinite set, and let $\mu(E)$ be the number of points in E if E is finite. This μ is called the **counting measure** on X .
2. Fix $x_0 \in X$, define $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$, for any $E \subset X$. This μ may be called the **unit mass concentrated at x_0** .
3. Let μ be the counting measure on the set $\{1, 2, 3, \dots\}$, let $A_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap A_n = \emptyset$ but $\mu(A_n) = \infty$ for $n = 1, 2, 3, \dots$. This shows that the hypothesis

$$\mu(A_1) < \infty$$

is not superfluous in Theorem 1.3.2(5).

1.4 Integration of Positive Functions

Definition 1.4.1 ▶ Integration of Positive Functions

1. If $s : X \rightarrow [0, \infty)$ is a measurable simple function, of the form

$$s = \sum_{i=1}^n \alpha_i \chi_{A_i},$$

where $\alpha_1, \dots, \alpha_n$ are the distinct values of s , and if $E \in \mathfrak{M}$, we define

$$\int_E s \, d\mu = \sum_{i=1}^n \alpha_i \mu(A_i \cap E).$$

The convention $0 \cdot \infty = 0$ is used here; it may happen that $\alpha_i = 0$ for some i and that $\mu(A_i \cap E) = \infty$.

2. If $f : X \rightarrow [0, \infty]$ is measurable, and $E \in \mathfrak{M}$, we define

$$\int_E f \, d\mu = \sup \int_E s \, d\mu,$$

the supremum being taken over all simple measurable functions s such that $0 \leq s \leq f$. The left member above is called the **Lebesgue integral** of f over E , with respect to the measure μ . It is a number in $[0, \infty]$.

Remark.

We apparently have two definitions for $\int_E f \, d\mu$ if f is simple, they are the same.

Corollary 1.4.2

- (a) If $0 \leq f \leq g$, then $\int_E f \, d\mu \leq \int_E g \, d\mu$.
- (b) If $A \subset B$ and $f \geq 0$, then $\int_A f \, d\mu \leq \int_B f \, d\mu$.
- (c) If $f \geq 0$ and c is a constant, $0 \leq c < \infty$, then

$$\int_E cf \, d\mu = c \int_E f \, d\mu.$$

- (d) If $f(x) = 0$ for all $x \in E$, then $\int_E f \, d\mu = 0$, even if $\mu(E) = \infty$.
- (e) If $\mu(E) = 0$, then $\int_E f \, d\mu = 0$, even if $f(x) = \infty$ for every $x \in E$.
- (f) If $f \geq 0$, then $\int_E f \, d\mu = \int_X \chi_E f \, d\mu$.

Remark.

We can also regard the $\int_E f \, d\mu$ as the restricted function $f|_E$ integrates on the induced measure space E .

Proposition 1.4.3

Let s and t be nonnegative measurable simple functions on X . For $E \in \mathfrak{M}$, define

$$\varphi(E) = \int_E s \, d\mu.$$

Then φ is a measure on $\mathfrak{M}$. Also

$$\int_X (s + t) \, d\mu = \int_X s \, d\mu + \int_X t \, d\mu.$$

Note.

$\varphi(E) = \int_E s \, d\mu$ 就是依据 s 对测度 μ 进行一个有限的算数的加权. 自然地也就成为一个测度.

Proof. If $E_1, E_2, \dots$ are disjoint members of $\mathfrak{M}$ whose union is E , the countable additivity of μ shows that

$$\begin{aligned} \varphi(E) &= \sum_{i=1}^n \alpha_i \mu(A_i \cap E) \\ &= \sum_{i=1}^n \sum_{j=0}^{\infty} \alpha_i \mu(A_i \cap E_j) \\ &= \sum_{j=0}^{\infty} \sum_{i=1}^n \alpha_i \mu(A_i \cap E_j) \\ &= \sum_{j=0}^{\infty} \varphi(E_j) \end{aligned}$$

Also, $\varphi(\emptyset) \neq \infty$, μ is not identically ∞ .

Next, let $\beta_1, \dots, \beta_m$ be distinct values of t . And let $B_j := \{x : t(x) = \beta_j\}$.

Then

$$\begin{aligned}
 \int_X (s + t) \, d\mu &= \sum_{i,j} \int_{A_i \cap B_j} (s + t) \, d\mu \\
 &= \sum_{i,j} \alpha_i \mu(A_i \cap B_j) + \sum_{i,j} \beta_j \mu(A_i \cap B_j) \\
 &= \sum_i \alpha_i \mu(A_i) + \sum_j \beta_j \mu(B_j) \\
 &= \int_X s \, d\mu + \int_X t \, d\mu
 \end{aligned}$$

□

Theorem 1.4.4 ▶ Lebesgue's Monotone Convergence Theorem

Let $\{f_n\}$ be a sequence of measurable functions on X , and suppose that

(a) $0 \leq f_1(x) \leq f_2(x) \leq \dots \leq \infty$ for every $x \in X$,

(b) $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, for every $x \in X$.

Then f is measurable, and

$$\int_X f_n \, d\mu \rightarrow \int_X f \, d\mu \quad \text{as } n \rightarrow \infty.$$

Proof. Since $\int_X f_n \, d\mu \leq \int_X f_{n+1} \, d\mu$ for all n , there exists an $\alpha > 0$ such that

$$\int_X f_n \, d\mu \rightarrow \alpha, \quad \text{as } n \rightarrow \infty$$

By Theorem 1.1.7, f is measurable, then by $\int_X f_n \, d\mu \leq \int_X f \, d\mu \, \forall n$,

$$\alpha \leq \int_X f \, d\mu$$

Take any simple function $0 \leq s \leq f$, and $0 \leq c < 1$. Define

$$E_n^s := \{x \in X : f_n(x) \geq cs(x)\}$$

Then since $\varphi(E) := \int_E s \, d\mu$ is a measure,

$$\alpha \geq \lim_{n \rightarrow \infty} \int_X f_n \, d\mu \geq \lim_{n \rightarrow \infty} \int_{E_n^s} f_n \, d\mu \geq c \lim_{n \rightarrow \infty} \int_{E_n^s} s \, d\mu = c \lim_{n \rightarrow \infty} \varphi(E_n^s) = c\varphi(X)$$

That is

$$\alpha \geq c \int_X s \, d\mu$$

Since s is arbitrary,

$$\alpha \geq c \int_X f \, d\mu$$

and since c is arbitrary ,

$$\alpha \geq \int_X f \, d\mu$$

which completes the proof. □

Corollary 1.4.5

Let $f, g : X \rightarrow [0, \infty]$ be measurable functions on X . Then $f + g$ is measurable and

$$\int_X (f + g) \, d\mu = \int_X f \, d\mu + \int_X g \, d\mu$$

Proof. Let $\{s_n\}, \{s'_n\}$ be simple function sequences in Theorem 1.2.2 of f, g , respectively . Then by the theorem above and the Proposition 1.4.3

$$\begin{aligned} \int_X (f + g) \, d\mu &= \lim_{n \rightarrow \infty} \int_X (s_n + s'_n) \, d\mu = \lim_{n \rightarrow \infty} \int_X s_n \, d\mu + \lim_{n \rightarrow \infty} \int_X s'_n \, d\mu \\ &= \int_X f \, d\mu + \int_X g \, d\mu \end{aligned}$$

□

Corollary 1.4.6

If $f_n : X \rightarrow [0, \infty]$ is **measurable**, for $n = 1, 2, 3, \dots$, and

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (x \in X),$$

then

$$\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu.$$

Proof. Let $g_n(x) = \sum_{k=1}^n f_k(x)$. Then $g_1 \leq g_2 \leq \dots \leq \infty$ and $f = \lim_{n \rightarrow \infty} g_n(x)$. Applying Monotone Convergence Theorem to $\{g_n\}$ and f , we have

$$\int_X f \, d\mu = \lim_{n \rightarrow \infty} \int_X \sum_{k=1}^n f_k(x) \, d\mu = \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_X f_k(x) \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu$$

□

Corollary 1.4.7

If $a_{ij} \geq 0$ for i and $j = 1, 2, 3, \dots$, then

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}.$$

Proof. Let μ be counting measure on $\mathbb{Z}$. Let

$$f_i(j) = a_{ij}, \quad f(k) = \sum_{i=1}^{\infty} f_i(k) = \sum_{i=1}^{\infty} a_{ik}$$

By Monotone Convergence Theorem, for any measurable g ,

$$\int_{\mathbb{Z}} g \, d\mu = \int_{\mathbb{Z}} \lim_{n \rightarrow \infty} \chi(\{1, \dots, n\}) g \, d\mu = \lim_{n \rightarrow \infty} \int_{\mathbb{Z}} \chi(\{1, \dots, n\}) g \, d\mu = \sum_{k=1}^n g(k)$$

Thus

$$\int_{\mathbb{Z}} f \, d\mu = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

And we have

$$\sum_{i=1}^{\infty} \int_X f_i \, d\mu = \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij}$$

The above corollary shows that

$$\int_{\mathbb{Z}} f \, d\mu = \sum_{i=1}^{\infty} \int_X f_i \, d\mu$$

That is

$$\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} a_{ij} = \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} a_{ij}$$

□

Lemma 1.4.8 ▶ Fatou's Lemma

If $f_n : X \rightarrow [0, \infty]$ is **measurable**, for each positive integer n , then

$$\int_X \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu \leq \liminf_{n \rightarrow \infty} \int_X f_n d\mu. \quad (1)$$

Note.

有了 Monotone Convergence Theorem 之后, 关键的不等式就剩下

$$\int_X \inf_{n \geq k} f_n \, d\mu \leq \inf_{n \geq k} \int_X f_n \, d\mu$$

它的含义是局部最小的积分小于等于积分的最小。

Proof. Let

$$g_k(x) = \inf_{n \geq k} f_n(x)$$

Then

$$\liminf_{n \rightarrow \infty} f_n = \lim_{n \rightarrow \infty} g_n =: g$$

where $\{g_k\}, g$ are all measurable. It is obvious that $\{g_k\}$ is increasing. Thus

$$\lim_{n \rightarrow \infty} \int_X g_n \, d\mu = \int_X g \, d\mu = \int_X \left(\liminf_{n \rightarrow \infty} f_n \right) d\mu$$

Only need to show that

$$\int_X \inf_{n \geq k} f_n \, d\mu \leq \inf_{n \geq k} \int_X f_n \, d\mu$$

which is true since

$$\int_X \inf_{n \geq k} f_n d\mu \leq \int_X f_n d\mu, \text{ for all } n \geq k$$

□

Theorem 1.4.9

Suppose $f : X \rightarrow [0, \infty]$ is measurable, and

$$\varphi(E) = \int_E f d\mu \quad (E \in \mathfrak{M}).$$

Then φ is a measure on $\mathfrak{M}$, and

$$\int_X g d\varphi = \int_X gf d\mu$$

for every measurable g on X with range in $[0, \infty]$.

Note.

将 measurable function f 视为对 measure μ 的某种 countable 次算数加权操作. 那么得益于 μ 对 countable 操作的相容性, φ 也就成为一个测度. φ 的积分行为也验证了这种看法的合理性.

Proof. • Let $E_1, \dots, E_n, \dots$ be disjoint measurable sets such that $E = \bigcup_{n=1}^{\infty} E_n$. Then $\{\sum_{k=1}^n \chi_{E_k} f\}$ be increasing measurable function sequence such that $\lim_{n \rightarrow \infty} \sum_{k=1}^n \chi_{E_k} f = f$. The Monotone Convergence Theorem shows that

$$\begin{aligned} \sum_{n=1}^{\infty} \varphi(E_n) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \int_{E_k} f d\mu = \lim_{n \rightarrow \infty} \int_E \sum_{k=1}^n \chi_{E_k} f d\mu \\ &= \int_E f d\mu = \varphi(E) \end{aligned}$$

Since $\varphi(\emptyset) = 0$. φ is a measure on X .

- To show the second results. See that it is true for $g = \chi_E$. Thus it is true for g is a simple function. And the general case follows from the

Monotone Convergence Theorem.



1.5 Integration of Complex Functions

Definition 1.5.1 ▶ Lebesgue Integrability

We define $L^1(\mu)$ to be the collection of all complex measurable functions f on X for which

$$\int_X |f| d\mu < \infty.$$

Note that the measurability of f implies that of $|f|$, hence the above integral is defined.

The members of $L^1(\mu)$ are called **Lebesgue integrable functions** (with respect to μ) or **summable functions**.

Definition 1.5.2 ▶ Complex Integration

If $f = u + iv$, where u and v are real measurable functions on X , and if $f \in L^1(\mu)$, we define

$$\int_E f d\mu = \int_E u^+ d\mu - \int_E u^- d\mu + i \int_E v^+ d\mu - i \int_E v^- d\mu \quad (1)$$

for every measurable set E .

Remark.

We have $u^+ \leq |u| < |f|$, etc., so that each of these four integrals is finite. Thus (1) defines the integral on the left as a complex number.

Definition 1.5.3 ▶ Real Integration

Occasionally it is desirable to define the integral of a measurable function f with range in $[-\infty, \infty]$ to be

$$\int_E f d\mu = \int_E f^+ d\mu - \int_E f^- d\mu, \quad (2)$$

provided that at least one of the integrals on the right of (2) is finite. The

left side of (2) is then a number in $[-\infty, \infty]$.

Theorem 1.5.4

Suppose f and $g \in L^1(\mu)$ and α and β are complex numbers. Then $\alpha f + \beta g \in L^1(\mu)$, and

$$\int_X (\alpha f + \beta g) d\mu = \alpha \int_X f d\mu + \beta \int_X g d\mu$$

Proof sketch.

可积性来自于模长的三角不等式. 主要证明可加性. 将 $f + g = h$ 的正负部分到两边, 就都变成了非负函数. 积分即可.

Theorem 1.5.5

If $f \in L^1(\mu)$, then

$$\left| \int_X f d\mu \right| \leq \int_X |f| d\mu.$$

Proof. Let $z = \int_X f d\mu$, z is a complex number. There exists a complex number α ($\alpha = \frac{\bar{z}}{|z|}$), such that $|\alpha| = 1$, and

$$\left| \int_X f d\mu \right| = \alpha z = \alpha \int_X f d\mu = \int_X \alpha f d\mu$$

Suppose $\alpha f = u + iv$, since $\int_X \alpha f d\mu$ is a real number, $\int_X v d\mu = 0$. We have

$$\int_X \alpha f d\mu = \int_X u d\mu \leq \int_X |\alpha f| d\mu = \int_X |f| d\mu$$

□

Theorem 1.5.6 ▶ Lebesgue's Dominated Convergence Theorem

Suppose $\{f_n\}$ is a sequence of complex measurable functions on X such that

$$f(x) = \lim_{n \rightarrow \infty} f_n(x)$$

exists for every $x \in X$. If there is a function $g \in L^1(\mu)$ such that

$$|f_n(x)| \leq g(x) \quad (n = 1, 2, 3, \dots; x \in X),$$

then $f \in L^1(\mu)$,

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| d\mu = 0,$$

and

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

Proof sketch.

$\{f_n\}$ 不一定单调, 能将极限从积分外塞进积分里的工具, 目前我们只有 Fatou's Lemma, 但是它原始的样子跟我们需要的不等号是反向的. 自然的想法是填一个负号, 这又会导致函数失去非负性, 而控制函数 g 恰好能弥补这一点不足. 实现起来, 我们从 g 上扣去 f, f_n , 而非直接拿出 f, f_n 去做.

Proof. Since $|f| \leq g$ and f is measurable, $f \in L^1(\mu)$. Function $(2g - |f_n - f|)$ is a nonnegative measurable function. Applying Fatou's Lemma to it, we have

$$\int_X \liminf_{n \rightarrow \infty} (2g - |f_n - f|) d\mu \leq \liminf_{n \rightarrow \infty} \int_X (2g - |f_n - f|) d\mu$$

where

$$\liminf_{n \rightarrow \infty} (2g - |f_n - f|) = 2g - \limsup_{n \rightarrow \infty} |f_n - f| = 2g$$

$$\liminf_{n \rightarrow \infty} \int_X (2g - |f_n - f|) d\mu = 2 \int_X g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu$$

The original inequality comes to

$$2 \int_X g d\mu \leq 2 \int_X g d\mu - \limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu$$

Since $\int_X g d\mu$ is finite, we can subtract $2 \int_X g d\mu$ both sides and get

$$\limsup_{n \rightarrow \infty} \int_X |f_n - f| d\mu \leq 0$$

Thus

$$\lim_{n \rightarrow \infty} \int_X |f_n - f| \, d\mu = 0$$

Finally, we have

$$\begin{aligned} \left| \lim_{n \rightarrow \infty} \int_X f_n \, d\mu - \int_X f \, d\mu \right| &= \left| \lim_{n \rightarrow \infty} \int_X (f_n - f) \, d\mu \right| \\ &\leq \lim_{n \rightarrow \infty} \int_X |f_n - f| \, d\mu \\ &= 0 \end{aligned}$$

which completes the proof. □

1.6 The Sets of Measure Zero

Definition 1.6.1

Let P be a property which a point x may or may not have. If μ is a measure on a σ -algebra $\mathfrak{M}$ and if $E \in \mathfrak{M}$, the statement “ P holds **almost everywhere on E** ” (abbreviated to “ P holds a.e. on E ”) means that there exists an $N \in \mathfrak{M}$ such that $\mu(N) = 0$, $N \subset E$, and P holds at every point of $E - N$.

Remark.

This concept of a.e. depends of course very strongly on the given measure, and we shall write “a.e. $[\mu]$ ” whenever clarity requires that the measure be indicated.

Example 1.6.2

For example, if f and g are measurable functions and if

$$\mu(\{x : f(x) \neq g(x)\}) = 0,$$

we say that $f = g$ a.e. $[\mu]$ on X , and we may write $f \sim g$.

Remark.

This is easily seen to be an equivalence relation. The transitivity ($f \sim g$ and $g \sim h$ implies $f \sim h$) is a consequence of the fact that the union of

two sets of measure 0 has measure 0.

Proposition 1.6.3 ► Completion of a Measure

If $f \sim g$, then, for every $E \in \mathfrak{M}$,

$$\int_E f d\mu = \int_E g d\mu.$$

Theorem 1.6.4

Let $(X, \mathfrak{M}, \mu)$ be a measure space, let $\mathfrak{M}^*$ be the collection of all $E \subset X$ for which there exist sets A and $B \in \mathfrak{M}$ such that $A \subset E \subset B$ and $\mu(B - A) = 0$, and define $\mu(E) = \mu(A)$ in this situation. Then $\mathfrak{M}^*$ is a σ -algebra, and μ is a measure on $\mathfrak{M}^*$.

Remark.

This extended measure μ is called **complete**, since all subsets of sets of measure 0 are now measurable; the σ -algebra $\mathfrak{M}^*$ is called the **μ -completion** of $\mathfrak{M}$. The theorem says that every measure can be completed, so, whenever it is convenient, we may assume that any given measure is complete; this just gives us more measurable sets, hence more measurable functions.

Proof. We begin by checking that μ is well defined for every $E \in \mathfrak{M}^*$. If for $i = 1, 2$, $A_i \subseteq E \subseteq B_i$, where $A_i, B_i \in \mathfrak{M}$, $\mu(B_i - A_i) = 0$. We need to show that $\mu(A_1) = \mu(A_2)$. Since $A_1 \subseteq E \subseteq B_2$,

$$\mu(A_1) \leq \mu(B_2) = \mu(A_2) + \mu(B_2 - A_2) = \mu(A_2)$$

By symmetric,

$$\mu(A_2) \leq \mu(A_1)$$

Thus $\mu(A_1) = \mu(A_2)$, $\mu(E)$ is well defined. Next, let us verify that $\mathfrak{M}^*$ has the three defining properties of a σ -algebra.

1. It is clearly that $X \in \mathfrak{M}^*$.
2. If $A \subseteq E \subseteq B$, then $B^c \subseteq E^c \subseteq A^c$. And since $\mu(A^c - B^c) = \mu(A^c \cap B) = \mu(B - A) = 0$, $E^c \in \mathfrak{M}^*$.
3. Suppose $A_i \subseteq E_i \subseteq B_i$, $i = 1, 2, \dots$, where $A_i, B_i \in \mathfrak{M}$, $\mu(B_i - A_i) = 0$.

Let $E = \bigcup E_i, A = \bigcup A_i, B = \bigcup B_i$. Since

$$B - A = \bigcup_i (B_i - A) \subseteq \bigcup_i (B_i - A_i)$$

$$\mu(B - A) \leq \sum_i \mu(B_i - A_i) = 0$$

Then by $A \subseteq E \subseteq B$, it follows that $E \in \mathfrak{M}^*$.

Finally, we show the countable additive of μ . If the above sets E_i are disjoint, the same is true of the sets A_i , then

$$\mu(E) = \mu(A) = \sum_i \mu(A_i) = \sum_i \mu(E_i)$$

□

Definition 1.6.5 ► More Measurable Functions

We say a function f defined on a set $E \in \mathfrak{M}$ is **measurable on X** . If $\mu(E^c) = 0$ and if $f^{-1}(V) \cap E$ is a measurable for every open set V .

Remark.

- If we define $f(x) = 0$ for $x \in E^c$, we obtain a measurable function on X , in the old sense.
- If our measure happens to be complete, we can define f on E^c in a perfectly arbitrary manner, and we still get a measurable function. The integral of f over any set $A \in \mathfrak{M}$ is independent of the definition of f on E^c ; therefore this definition need not even be specified at all.

Theorem 1.6.6

Suppose $\{f_n\}$ is a sequence of complex measurable functions defined a.e. on X such that

$$\sum_{n=1}^{\infty} \int_X |f_n| d\mu < \infty. \quad (1)$$

Then the series

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad (2)$$

converges for almost all x , $f \in L^1(\mu)$, and

$$\int_X f d\mu = \sum_{n=1}^{\infty} \int_X f_n d\mu. \quad (3)$$

Remark.

- 现在, 考察积分和求和是否可交换, 只需要检查模长积分的级数是否收敛.
- 即便 f_n 在 X 上处处被定义, 级数收敛的地方也只是 ***almost everywhere***.

Note.

此时, 控制函数就是绝对级数本身, 它在一个满测集上将原级数控制住. 其特殊性间接给出了原级数的几乎处处收敛性质.

Proof. Let S_n be the set on which f_n is defined, so that $\mu(S_n^c) = 0$. Let $S = \bigcap S_n$. Put $\varphi(x) = \sum |f_n(x)|$, for $x \in S$. By Monotone Convergence Theorem

$$\int_S \varphi d\mu \leq \sum_{n=1}^{\infty} \int_S |f_n| d\mu < \infty$$

Let $E = \{x \in S : \varphi(x) < \infty\}$. Then it follows that $\mu(E^c) = 0$. We have

$$\sum_{n=1}^{\infty} |f_n(x)| < \infty, \quad \forall x \in E$$

Thus

$$f(x) = \sum_{n=1}^{\infty} f_n(x)$$

converges on E , which is a set with its complement measure zero. That is to say f converges for almost all x . And since $f \in L^1(\mu|_E)$, $\mu(E) = 0$, we have

$f \in L^1(\mu)$. Finally, by Monotone Convergence Theorem,

$$\int_X f \, d\mu = \int_E f \, d\mu = \sum_{n=1}^{\infty} \int_E f_n \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu$$

□

Theorem 1.6.7

1. Suppose $f : X \rightarrow [0, \infty]$ is measurable, $E \in \mathfrak{M}$, and $\int_E f \, d\mu = 0$. Then $f = 0$ a.e. on E .
2. Suppose $f \in L^1(\mu)$ and $\int_E f \, d\mu = 0$ for every $E \in \mathfrak{M}$. Then $f = 0$ a.e. on X .
3. Suppose $f \in L^1(\mu)$ and

$$\left| \int_X f \, d\mu \right| = \int_X |f| \, d\mu$$

Then there is a constant α such that $\alpha f = |f|$ a.e. on X .

Proof. 1. Let $A_n = \left\{ x \in E : f(x) \geq \frac{1}{n} \right\}$. Then

$$0 = \int_E f \, d\mu \geq \int_{A_n} f \, d\mu \geq \mu(A_n) \frac{1}{n}.$$

It follows that

$$\mu(A_n) = 0, \quad \forall n = 1, 2, \dots$$

Let $A = \bigcup_{n=1}^{\infty} A_n$. Then $\mu(A) = 0$, and $f \equiv 0$ on $E \setminus A$. Thus $f = 0$ a.e. on E .

2. Suppose $f = u + iv$. Let $E = \{x \in X : u(x) \geq 0\}$. Since $\operatorname{Re} f = u^+$ on E , then

$$\int_E u^+ \, d\mu = 0$$

It follows by 1. that $\mu^+ = 0$ a.e. on E . Since $\mu^+(X \setminus E) \equiv 0$, $\mu^+ = 0$ a.e. on X . We conclude similarly that

$$u^- = v^+ = v^- = 0, \quad a.e.$$

□